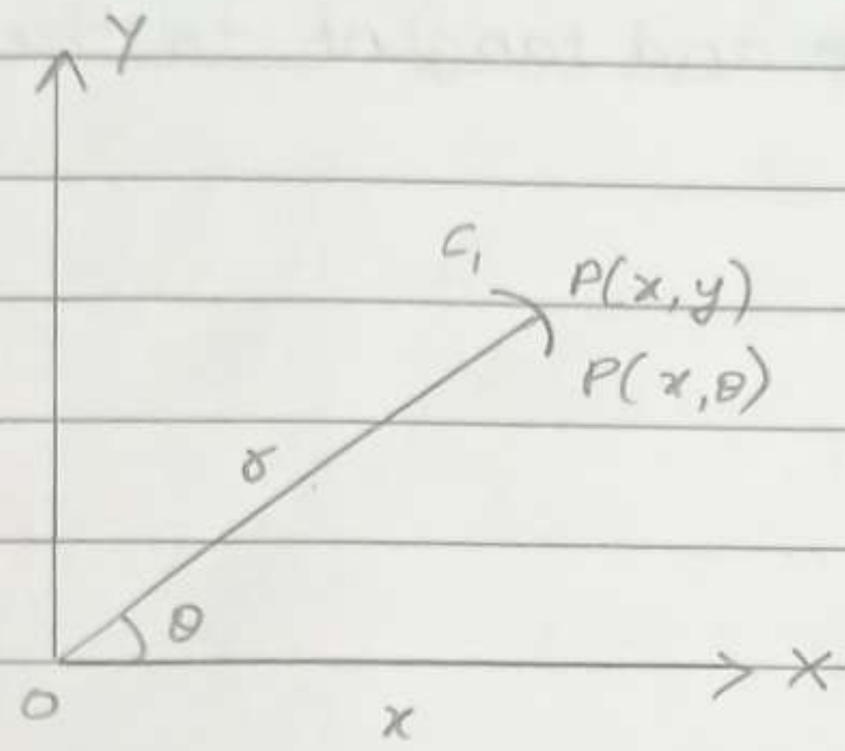
classmate
Date 05-02-25
Page

Differential Calculus

Polar Curve



Let 'P' be any point on the x-y plane. Join O and P. Let 'r' be the length of line segment OP and 'θ' be the angle between OP and +ve X-axis (initial line). Then (r, θ) represents the polar co-ordinates at a point P. Let (x, y) be the cartesian co-ordinates at a point P. We have P(x, y) = P(r, θ). From the graph, we have, $\cos\theta = \frac{adj}{hypo.} = \frac{x}{r}$ and $\sin\theta = \frac{opp}{hyp} = \frac{y}{r}$

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

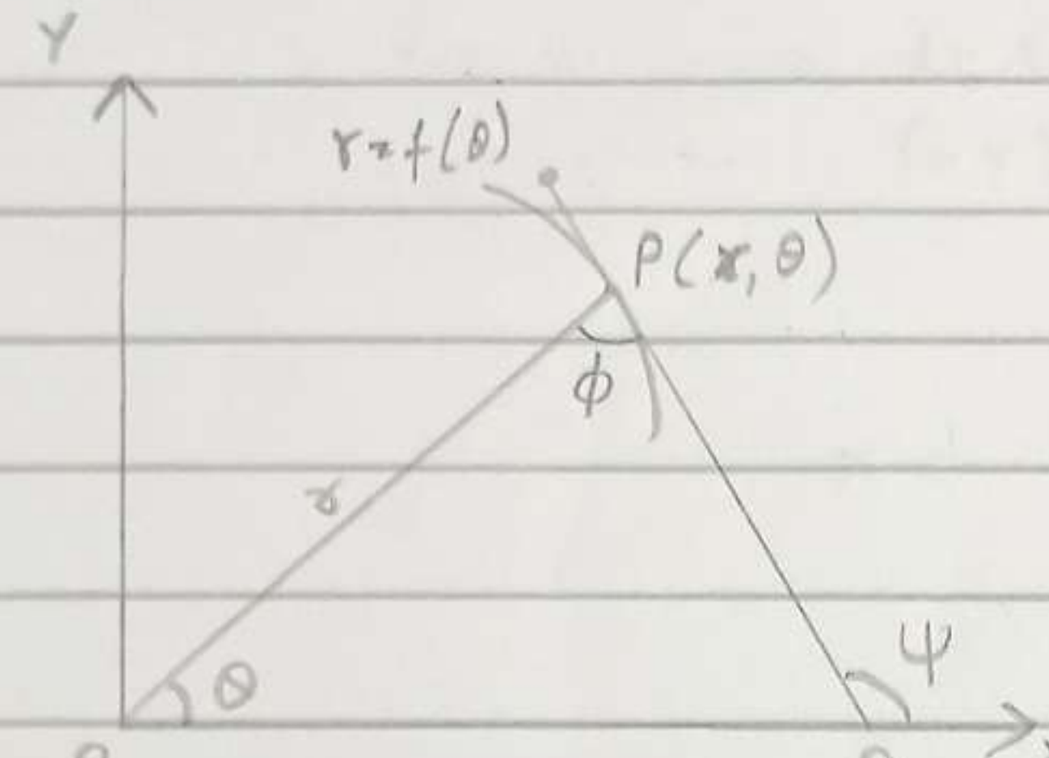
Taking,

$$\frac{\sin\theta}{\cos\theta} = \frac{y}{x} \Rightarrow \tan\theta = \frac{y}{x} \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

A curve specified by a polar equation, $r = f(\theta)$ is called 'polar curve'.

Imp derivation

Angle between radius vector and tangent to the polar curve



Let $P(r,\theta)$ be a point on the polar curve $r = f(\theta)$ and PQ be the tangent at a point P and joining it to the initial line (+ve x-axis) at Q.

Let ψ be the angle between the tangent line and the initial line. Let ϕ be the angle between radius vector and tangent line.

Then, by the trigonometric property, we have,

$$\psi = \theta + \phi$$

$$\Rightarrow \tan\psi = \tan(\theta + \phi) \qquad \because \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$\Rightarrow \tan\psi = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \cdot \tan\phi}$$

= f(θ) is

to the

f(θ) and
to the

ne and
n radius

ave,

$\frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$

We know that, slope of the tangent is $\frac{dy}{dx} = \tan\psi$

Hence, $\frac{dy}{dx} = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \cdot \tan\phi}$ ——— ①

W.K.T., $x = r\cos\theta$, $y = r\sin\theta$

$$\Rightarrow \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{d(r\sin\theta)/d\theta}{d(r\cos\theta)/d\theta}$$

$$\Rightarrow \frac{dy}{dx} = \frac{r\cos\theta + \sin\theta \cdot r'}{-r\sin\theta + \cos\theta \cdot r'}$$

$\frac{\div}{\div}$ by $r'\cos\theta$,

$$\frac{dy}{dx} = \frac{\frac{r\cos\theta}{r'\cos\theta} + \frac{r'\sin\theta}{r'\cos\theta}}{\frac{-r\sin\theta}{r'\cos\theta} + \frac{r'\cos\theta}{r'\cos\theta}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\frac{r}{r'} + \tan\theta}{\frac{-r}{r'}\tan\theta + 1} = \frac{\tan\theta + \frac{r}{r'}}{1 - \tan\theta\frac{r}{r'}}$$ ——— ②

Comparing eqn ① and ②, we get,

$$\tan\phi = f(\theta)\frac{r}{r'} \Rightarrow \tan\phi = \frac{r}{dr/d\theta}$$

$$\Rightarrow \tan\phi = r\frac{d\theta}{dr}$$

$$\Rightarrow \boxed{\cot\phi = \frac{1}{r}\frac{dr}{d\theta}}$$

Find the angle between the radius vector & tangent line.

1. $r^2 \cos 2\theta = a^2$

Soln: diff. w.r.t. θ on both sides,

Method 1°

$$r^2(-\sin 2\theta)2 + \cos 2\theta \cdot 2r\frac{dr}{d\theta} = 0$$

$$\Rightarrow -2r^2 \sin 2\theta + 2r\cos 2\theta \frac{dr}{d\theta} = 0$$

$$\Rightarrow 2r\cos 2\theta \frac{dr}{d\theta} = 2r^2 \sin 2\theta$$

$$\Rightarrow \frac{dr}{d\theta} = \frac{r\sin 2\theta}{\cos 2\theta}$$

$$\Rightarrow \frac{1}{r}\frac{dr}{d\theta} = \frac{\sin 2\theta}{\cos 2\theta}$$

$$\Rightarrow \cot\phi = \tan 2\theta \qquad \because \frac{1}{r}\frac{dr}{d\theta} = \cot\phi$$

$$\Rightarrow \cot\phi = \cot\left(\frac{\pi}{2} - 2\theta\right)$$

$$\Rightarrow \phi = \frac{\pi}{2} - 2\theta$$

=

2. $r = a(1-\cos\theta)$

Soln: Taking log on both sides,

Method 2°

$$\log r = \log(a(1-\cos\theta))$$

$$\Rightarrow \log r = \log a + \log(1-\cos\theta)$$

diff. w.r.t. θ,

$$\frac{1}{r}\frac{dr}{d\theta} = 0 + \frac{0+\sin\theta}{1-\cos\theta}$$

$$\Rightarrow \frac{1}{r}\frac{dr}{d\theta} = \frac{\sin\theta}{1-\cos\theta}$$

$\because \sin 2\theta = 2\sin\theta\cos\theta$

$\Rightarrow \sin\theta = 2\sin\theta/2 \cos\theta/2$

$\because 2\sin^2\theta = 1-\cos 2\theta$

$$\Rightarrow \cot\phi = \frac{2\sin\theta/2\cos\theta/2}{2\sin^2\theta/2} = \frac{\cos\theta/2}{\sin\theta/2}$$

$$\Rightarrow \cot\phi = \cot\theta/2$$

$$\Rightarrow \phi = \theta/2.$$

$=$

3. $\frac{2a}{r} = 1-\cos\theta$

Soln: Taking log on both sides,

$$\log\left(\frac{2a}{r}\right) = \log(1-\cos\theta)$$

$$\Rightarrow \log 2a - \log r = \log(1-\cos\theta)$$

$\Rightarrow$ diff. w.r.t. θ on both sides,

$$-\frac{1}{r}\frac{dr}{d\theta} = \frac{\sin\theta}{1-\cos\theta}$$

$$\Rightarrow \cot\phi = -\left(\frac{\cos\theta/2}{\sin\theta/2}\right) = -\cot\phi/2$$

$$\Rightarrow \cot\phi = -\cot\theta/2$$

$$\Rightarrow \phi = -\theta/2$$

$/$

4. $r^n = a^n \cos n\theta$

Soln: log on both sides,

$\log r^n = \log(a^n \cos n\theta)$

$\Rightarrow n \log r = \log a^n + \log \cos n\theta \quad = n\log a + \log \cos n\theta$

$\Rightarrow$ diff. w.r.t. θ on both sides,

$$\Rightarrow \frac{n}{r}\frac{dr}{d\theta} + 0 = \frac{-n \sin n\theta}{\cos n\theta}$$

$$\Rightarrow \frac{n}{r}\frac{dr}{d\theta} = -n \tan n\theta$$

$\Rightarrow \cot\phi = -\tan n\theta$

$\Rightarrow \cot\phi = +\cot\left(\frac{\pi}{2} + n\theta\right)$

$\Rightarrow \phi = n\theta + \frac{\pi}{2}$ //

5. $r = ae^{\theta \cot\alpha}$

Soln: log on both sides,

$\log r = \log ae^{\theta\cot\alpha}$

$\Rightarrow \log r = \log a + \theta\cot\alpha \log e$

$\Rightarrow \log r = \log a + \theta\cot\alpha$

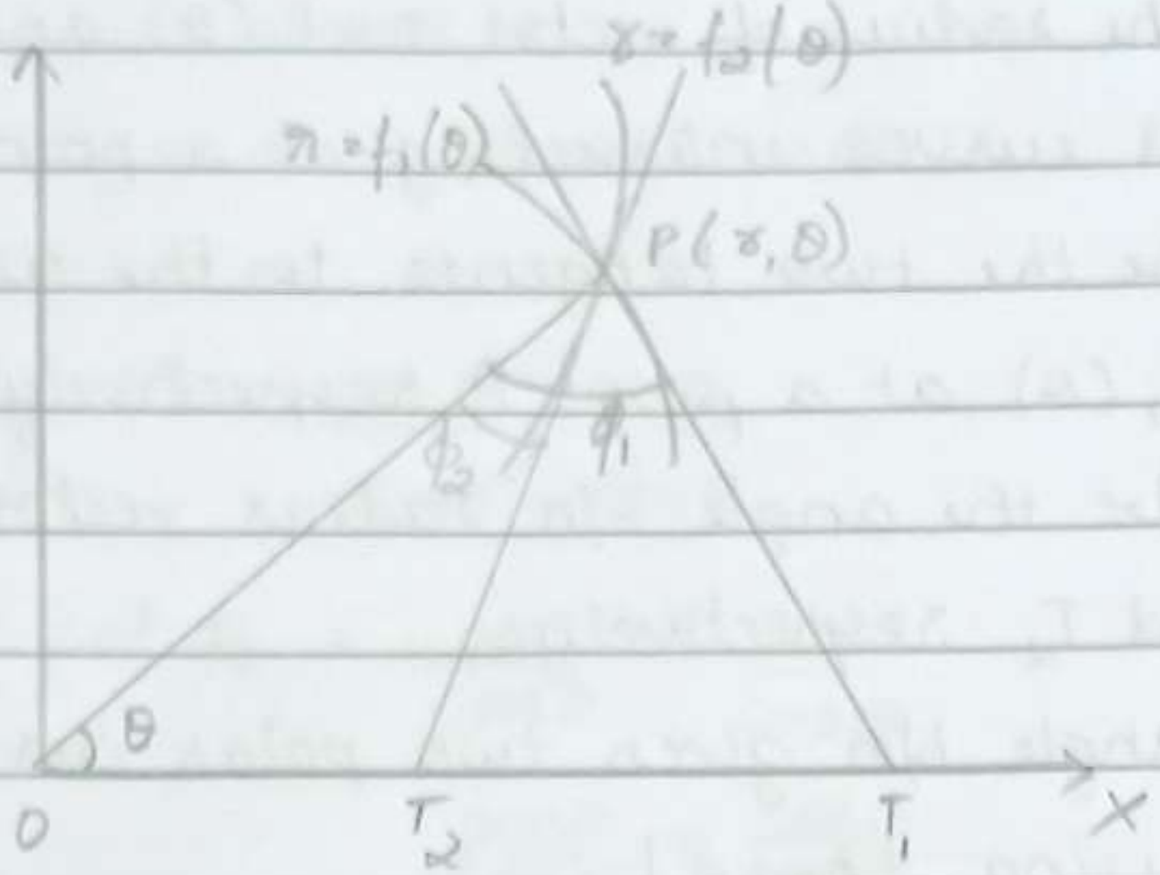
$\Rightarrow$ diff. w.r.t. θ,

$$\Rightarrow \frac{1}{r} r' = \cot\alpha$$

$\Rightarrow \cot\phi = \cot\alpha$

$\Rightarrow \phi = \alpha$

Angle between intersection of two curves



Note:

① $\sin(-\theta) = -\sin\theta$

$\cos(-\theta) = \cos\theta$

$-\tan n\theta = \cot(\pi/2 + n\theta)$

② $\sin(\pi/2 + \theta) = \cos\theta$

$\cos(\pi/2 + \theta) = -\sin\theta$

$\tan(\pi/2 + \theta) = -\cot\theta$

$\operatorname{cosec}(\pi/2 + \theta) = \sec\theta$

$\sec(\pi/2 + \theta) = -\operatorname{cosec}\theta$

$\cot(\pi/2 + \theta) = -\tan\theta$

③ $\sin(\pi/2 - \theta) = \cos\theta$

$\cos(\pi/2 - \theta) = \sin\theta$

$\tan(\pi/2 - \theta) = \cot\theta$

$\operatorname{cosec}(\pi/2 - \theta) = \sec\theta$

$\sec(\pi/2 - \theta) = \operatorname{cosec}\theta$

$\cot(\pi/2 - \theta) = \tan\theta$

$2\tan^{-1}(e) = \left| \frac{2e}{1 - e^2} \right|$

Let $OP = r$ be the radius of vector $r = f_1(\theta)$ and $r = f_2(\theta)$ be the two polar curves intersecting at a point P.
Let T_1 and T_2 be the two tangents to the curve $r = f_1(\theta)$ & $r = f_2(\theta)$ at a point P respectively.
Let ϕ_1 and ϕ_2 be the angle b/n radius vector and the tangents T_1 and T_2 respectively.
Therefore, the angle b/n given two polar curves can be evaluated using $|\phi_2 - \phi_1|$
If $|\phi_2 - \phi_1| = \pi/2$, then, the two curves intersect orthogonally.

$$\tan|\phi_2 - \phi_1| = \left| \frac{\tan\phi_2 - \tan\phi_1}{1 + \tan\phi_2 \tan\phi_1} \right|$$

① Find the angle b/n two polar curves.

a] $r = 2\sin\theta$ and $r = \sin\theta + \cos\theta$

LHS :

$r = 2\sin\theta$

$\Rightarrow \log r = \log 2 + \log\sin\theta$

diff. w.r.t. θ,

$\Rightarrow \frac{1}{r} r' = 0 + \frac{\cos\theta}{\sin\theta}$

$\Rightarrow \cot\phi_1 = \cot\theta$

$\Rightarrow \phi_1 = \theta$

RHS :

$r = \sin\theta + \cos\theta$

$\Rightarrow \log r = \log(\sin\theta + \cos\theta)$

diff. w.r.t. θ,

$$\frac{1}{r} r' = \frac{\cos\theta - \sin\theta}{\sin\theta + \cos\theta}$$

$$\Rightarrow \cot\phi_2 = \frac{\frac{\cos\theta}{\cos\theta} - \frac{\sin\theta}{\cos\theta}}{\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\cos\theta}}$$

$$= \frac{1 - \tan\theta}{\tan\theta + 1}$$

$$\Rightarrow \cot\phi_2 = \tan(\pi/4 - \theta)$$

$$= \cot(\pi/2 - (\pi/4 - \theta))$$

$$\Rightarrow \cot\phi_2 = \cot(\pi/2 + \theta) \Rightarrow \phi_2 = (\pi/4 + \theta)$$

$\Rightarrow$

$$|\phi_2 - \phi_1| = |\pi/4 + \theta - \theta| = \pi/4$$

b] $r = a\theta$ and $r = \frac{a}{\theta}$

Soln: LHS : $\log r = \log(a\theta)$

$\Rightarrow \log r = \log a + \log\theta$

diff. w.r.t. θ,

$$\frac{1}{r} r' = \frac{1}{\theta} \Rightarrow \cot\phi_1 = \frac{1}{\theta} \Rightarrow \tan\phi_1 = \theta \quad \text{---} ①$$

RHS : $r = \frac{a}{\theta}$

$\Rightarrow \log r = \log a - \log \theta$

$\Rightarrow$ diff. wrt θ,

$\Rightarrow \frac{1}{r} r' = \frac{-1}{\theta}$

$\Rightarrow \cot \phi_2 = \frac{-1}{\theta}$

$\Rightarrow \tan \phi_2 = -\theta$ ——— (2)

Since the two curves intersect at a point,

$r_1 \theta = \frac{a}{\theta}$

$\Rightarrow \theta^2 = 1$

$\Rightarrow \theta = \pm 1$

when $\theta = 1$,

$\tan \phi_1 = \theta$

$\Rightarrow \phi_1 = \tan^{-1} \theta$

$= \tan^{-1}(1)$

$= \pi/4$

$\Rightarrow \tan \phi_2 = -\theta$

$\Rightarrow \phi_2 = \tan^{-1}(-\theta) = \tan^{-1}(-1) = 3\pi/4$ or $(-\pi/4)$

$$|\phi_2 - \phi_1| = \left|\frac{3\pi}{4} - \frac{\pi}{4}\right| = \pi/2$$

When $\theta = -1$,

$\tan\phi_1 = \theta$

$\Rightarrow \phi_1 = \tan^{-1}\theta$

$\Rightarrow \phi_1 = \tan^{-1}(-1)$

$\Rightarrow \phi_1 = \frac{3\pi}{4}$ or $\left(\frac{-\pi}{4}\right)$

$\tan\phi_2 = -\theta$

$\Rightarrow \phi_2 = \tan^{-1}(-\theta) = \tan^{-1}(1) = \pi/4$

$$|\phi_2 - \phi_1| = \left|\pi/4 - \frac{3\pi}{4}\right| = \pi/2 =$$

c] $r = 3\cos\theta$ and $r = 1 + \cos\theta$.

Prob: LHS :

$\log r = \log 3\cos\theta$

$\Rightarrow \log r = \log 3 + \log\cos\theta$

$\Rightarrow \frac{1}{r} r' = \frac{-\sin\theta}{\cos\theta}$

$\Rightarrow \cot\phi_1 = -\tan\theta$

$\Rightarrow \cot\phi_1 = \cot(\pi/2 + \theta)$

$\Rightarrow \phi_1 = \pi/2 + \theta$.

Equating,

$3\cos\theta = 1 + \cos\theta$

$2\cos\theta = 1 \Rightarrow \cos\theta = 1/2$

$\Rightarrow \theta = \pi/3$

RHS:

$r = 1 + \cos\theta$

$\Rightarrow \log r = \log(1 + \cos\theta)$

$\Rightarrow \log r = \log(1 + \log\cos\theta)$

$\Rightarrow \frac{1}{r} r' = \frac{-\sin\theta}{1+\cos\theta} = \frac{-2\sin\theta/2\cos\theta/2}{2\cos^2\theta/2} = -\tan\theta/2$

$\Rightarrow \cot\phi = -\tan\theta/2 = \cot\left(\frac{\pi}{2} + \frac{\theta}{2}\right)$

$\Rightarrow \phi_2 = \frac{\pi}{2} + \theta/2$

$\Rightarrow |\phi_2 - \phi_1| = \left|\frac{\pi}{2} + \theta - \frac{\pi}{2} - \theta/2\right| = \frac{\theta}{2} = \frac{\pi}{6}$

d] $r = a\log\theta$ and $r = \frac{a}{\log\theta}$

Soln: $\log r = \log a + \log\theta$

$\Rightarrow \frac{1}{r} r' = 0 + \frac{1}{\log\theta} \cdot \frac{1}{\theta}$

$\Rightarrow \cot\phi_1 = \frac{1}{\theta\log\theta}$

$\Rightarrow \tan\phi_1 = \theta\log\theta$

$\log r = \log a - \log(\log\theta)$

$\Rightarrow \frac{1}{r} r' = 0 - \frac{1}{\theta\log\theta}$

$\Rightarrow \cot\phi_2 = \frac{-1}{\theta\log\theta}$

$\Rightarrow \tan\phi_2 = -\theta\log\theta$

At the point of intersection,

$a\log\theta = \frac{a}{\log\theta}$

$\Rightarrow \log^2\theta = 1$

$\Rightarrow \log\theta = 1$

$\Rightarrow \theta = e$

Now,

$|\phi_2 - \phi_1| = |-\theta\log\theta - \theta\log\theta| = 2\theta\log\theta = 2e\log e = 2e$

$\Rightarrow \tan|\phi_2 - \phi_1| = \left|\frac{\tan\phi_2 - \tan\phi_1}{1+\tan\phi_1\tan\phi_2}\right| = \left|\frac{-\theta-\theta}{1-\theta^2}\right| = \left|\frac{-2\theta}{1-\theta^2}\right|$

$= \frac{2e}{e^2-1}$

e] $r = \frac{a\theta}{1+\theta}$ and $r = \frac{a}{1+\theta^2}$

Now $\log r = \log(a\theta) - \log(1+\theta) = \log a + \log\theta - \log(1+\theta)$

$\Rightarrow \frac{1}{r}r' = 0 + \frac{1}{\theta} - \frac{1}{(1+\theta)}$

$\Rightarrow \cot\phi_1 = \frac{1+\theta-\theta}{\theta(1+\theta)} = \frac{1}{\theta(1+\theta)} = \frac{1}{\theta+\theta^2}$ ——— ①

$$\log \gamma = \log a - \log(1+\theta^2)$$

$$\Rightarrow \frac{1}{\gamma}\gamma' = 0 - \frac{2\theta}{(1+\theta^2)}$$

$$\Rightarrow \cot\phi_2 = \frac{-2\theta}{(1+\theta^2)} \quad \text{——— (2)}$$

At point of intersection,

$$\frac{1}{\theta+\theta^2} = \frac{-2\theta}{1+\theta^2} \qquad \frac{a\theta}{1+\theta} = \frac{a}{1+\theta^2}$$

$$\Rightarrow 1+\theta^2 = -2\theta^2 \qquad \Rightarrow \theta(1+\theta^2) = 1+\theta$$

$$\Rightarrow \theta+\theta^3 = 1+\theta$$

$$\Rightarrow \theta^3 = 1$$

$$\Rightarrow \theta = 1$$

$$\cot\phi_1 = \frac{1}{\theta+\theta^2} = \frac{1}{1+1^2} = \frac{1}{2} \Rightarrow \tan\phi_1 = 2$$

$$\cot\phi_2 = \frac{-2\theta}{1+\theta^2} = \frac{-2(1)}{1+1} = \frac{-2}{2} = -1 \Rightarrow \tan\phi_2 = -1$$

Now,

$$\tan|\phi_1-\phi_2| = \left|\frac{\tan\phi_1-\tan\phi_2}{1+\tan\phi_1\tan\phi_2}\right| = \left|\frac{-1-2}{1+(-1)(2)}\right| = \frac{3}{1}$$

$$\Rightarrow \phi_1-\phi_2 = \tan^{-1}(3)$$

Q1. Proove that the following curves interseet orthogonally (angle is $\pi/2$).

$r = \frac{a}{1+\cos\theta}$; $r = \frac{b}{1-\cos\theta}$

Soln: $\log r = \log a - \log(1+\cos\theta)$

$\Rightarrow \frac{1}{r} r' = 0 + \frac{\sin\theta}{(1+\cos\theta)}$

$\Rightarrow \cot\phi_1 = \frac{\sin\theta}{1+\cos\theta} = \frac{2\sin\theta/2\cos\theta/2}{2\cos^2\theta/2} = \tan\theta/2$

$= \cot(\pi/2 - \theta/2)$,

$\Rightarrow \phi_1 = \frac{\pi}{2} - \theta/2$,

$\log r = \log b - \log(1-\cos\theta)$

$\Rightarrow \frac{1}{r} r' = 0 - \frac{\sin\theta}{1-\cos\theta}$

$\Rightarrow \cot\phi_2 = \frac{-\sin\theta}{1-\cos\theta} = \frac{-2\sin\theta/2\cos\theta/2}{2\sin^2\theta/2} = -\cot\theta/2 = \cot\theta/2$

$= \cot(\frac{\pi}{2} + \frac{\theta}{2})$

$\Rightarrow \phi_2 = -\theta/2$,

Now, $|\phi_2 - \phi_1| = |-\theta/2 - \frac{\pi}{2} + \theta/2| = \frac{\pi}{2}$,

Q2. $r = ae^\theta$ and $re^\theta = b$.

Soln: $\log r = \log a + \log e^\theta = \log a + \theta\log e = \log a + \theta$.

$\Rightarrow \frac{1}{r} r' = 1 \Rightarrow \cot\phi_1 = 1 \Rightarrow \phi_1 = \pi/4$,

$\log r = \log b - \log e^\theta = \log b - \theta\log e = \log b - \theta$

$\Rightarrow \frac{1}{r} r' = -1 \Rightarrow \cot\phi_2 = -1 \Rightarrow \phi_2 = -\pi/4$,

$$\Rightarrow |\phi_2 - \phi_1| \cdot \left|-\pi/4 - \pi/4\right| \cdot \frac{\pi}{2}$$

Q3 $r = a(1+\cos\theta)$; $r = b(1-\cos\theta)$

Soln: $\log r = \log a + \log(1+\cos\theta)$

$$\Rightarrow \cot\phi_1 = 0 + \frac{1}{(1+\cos\theta)}(-\sin\theta) = \frac{-2\sin\theta/2\cos\theta/2}{2\cos^2\theta/2} = -\tan\theta/2 = \cot(\pi/2 + \theta/2)$$

$$\Rightarrow \phi_1 = \pi/2 + \theta/2,$$

$$\log r = \log b + \log(1-\cos\theta)$$

$$\Rightarrow \cot\phi_2 = 0 + \frac{\sin\theta}{1-\cos\theta} = \frac{2\sin\theta/2\cos\theta/2}{2\sin^2\theta/2} = \cot\theta/2 \Rightarrow \phi_2 = \theta/2,$$

$$|\phi_2 - \phi_1| = |\theta/2 - \pi/2 - \theta/2| = \pi/2$$

Q4 $r^n = a^n\cos n\theta$; $r^n = b^n\sin\theta$

Soln: $\log r^n = \log a^n + \log(\cos n\theta)$

$$\Rightarrow \frac{1}{r}\cdot r' n = 0 + \frac{(-\sin n\theta)\, n}{\cos n\theta}$$

$$\Rightarrow \cot\phi_1 = \frac{-\sin n\theta}{\cos n\theta} = \cot(\pi/2 + n\theta) \Rightarrow \phi_1 = \pi/2 + n\theta,$$

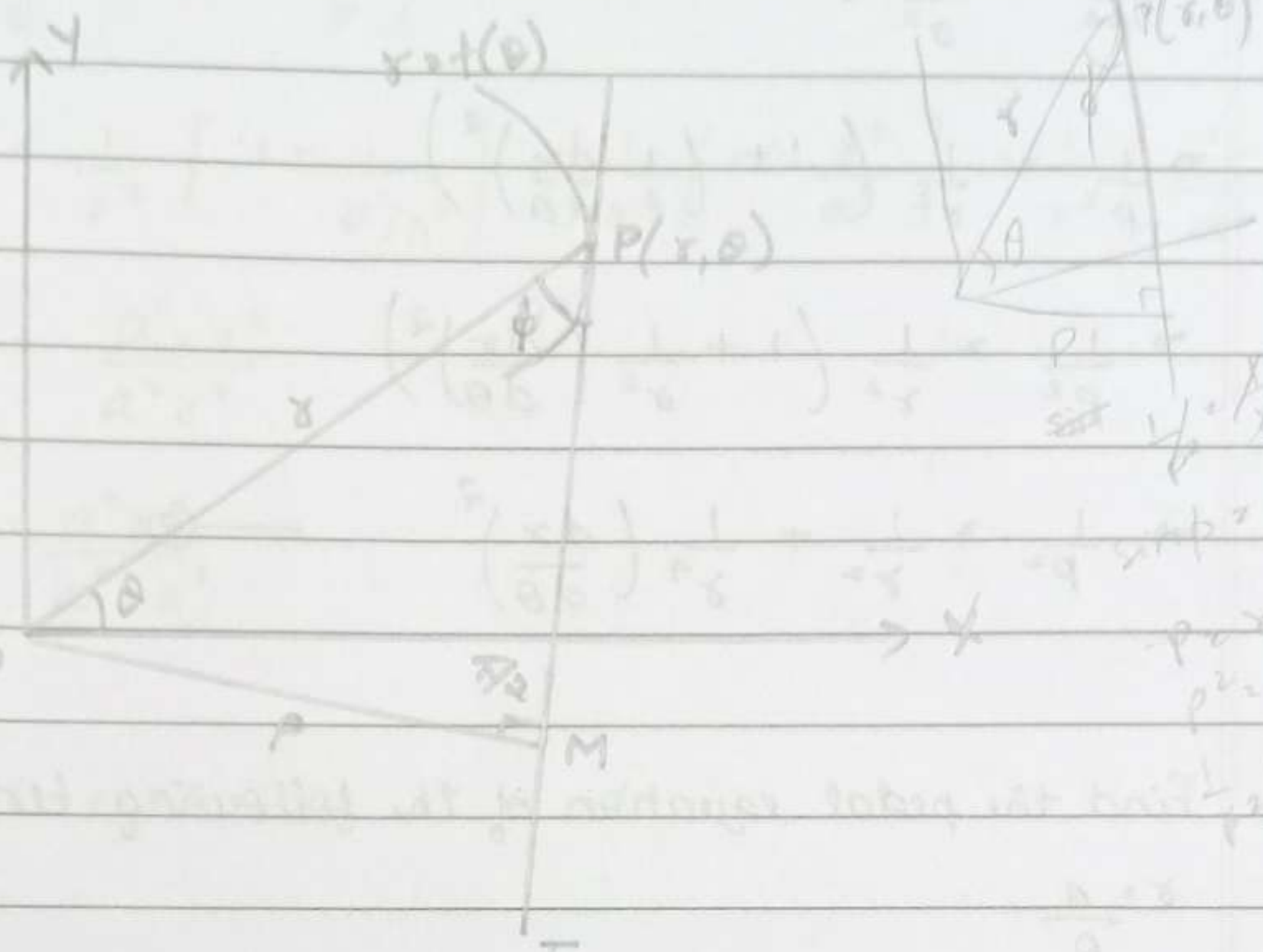
$$\log r^n = \log b^n \sin n\theta$$

$$\Rightarrow n\log r = \log b^n + \log\sin\theta \Rightarrow n\frac{1}{r}r' = \frac{\cos n\theta}{\sin n\theta}(n) \Rightarrow \cot\phi_2 = \cot n\theta$$

$$\Rightarrow \phi_2 = n\theta$$

$$|\phi_2 - \phi_1| = |n\theta - \pi/2 - n\theta| = \pi/2,$$

Pedal Equation



Poove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4}\left(\frac{dr}{d\theta}\right)^2$

Let $OP = r$ be the radius of vector. Let $r = f(\theta)$ be the polar curve. Let T be the tangent drawn at a point P with an angle of ϕ with radius vector and intersecting a line from centre O at M with an angle of $\pi/2$. Let $OM = p$ and angle between initial line and radius vector is θ.

From the three angle $\angle OPM$, we have,

$$\sin\phi = \frac{p}{r} \Rightarrow p = r\sin\phi \text{ -(1)} \Rightarrow p^2 = r^2\sin^2\phi$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\frac{1}{\sin^2\phi} \qquad \Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\operatorname{cosec}^2\phi$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\left(1+\cot^2\phi\right) \quad \text{—(2)}$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\left(1+\left(\frac{1}{r}\frac{dr}{d\theta}\right)^2\right)$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\left(1+\frac{1}{r^2}\left(\frac{dr}{d\theta}\right)^2\right)$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}+\frac{1}{r^4}\left(\frac{dr}{d\theta}\right)^2 \quad \text{—(3)}$$

Q1 Find the pedal equation of the following term.

$r = \frac{a}{\theta}$.

Soln: $\log r = \log a - \log\theta$

$$\Rightarrow \frac{1}{r}r' = 0 - \frac{1}{\theta}$$

$$\Rightarrow \cot\phi = \frac{1}{\theta} \quad \text{—(1)}$$

W.K.T., $\frac{1}{p^2} = \frac{1}{r^2}\left[1+\cot^2\phi\right]$

$$= \frac{1}{r^2}\left[1+(\cot\phi)^2\right]$$

$$= \frac{1}{r^2}\left[1+\left(\frac{-1}{\theta}\right)^2\right]$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\left[1+\frac{1}{\theta^2}\right]$$

$$\left[r = \frac{a}{\theta} \Rightarrow \theta = \frac{a}{r} \right]$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\left[1 + \frac{1}{a^2/r^2}\right] = \frac{1}{r^2}\left[1 + \frac{r^2}{a^2}\right] = \frac{1}{r^2}\left[\frac{a^2+r^2}{a^2}\right]$$

$$\Rightarrow \frac{1}{p^2} = \frac{a^2+r^2}{a^2 r^2}$$

$$\Rightarrow p^2 = \frac{a^2 r^2}{a^2+r^2}$$

Q2 $\frac{2a}{r} = 1 - \cos\theta$

$$\Rightarrow r = \frac{2a}{1-\cos\theta}$$

$$\Rightarrow \log r = \log 2a - \log(1-\cos\theta)$$

$$\Rightarrow \frac{1}{r} r' = 0 - \frac{\sin\theta}{(1-\cos\theta)} = \frac{-2\sin\theta/2\cos\theta/2}{2\sin^2\theta/2} = -\cot\theta/2$$

$$\Rightarrow \cot\phi = -\cot\theta/2 \quad \Rightarrow \phi = -\theta/2$$

WKT., $\frac{1}{p^2} = \frac{p}{r\sin\phi} = r\sin -\theta/2$

$$\frac{1}{p^2} = \frac{1}{r^2}\left(1 + (\cot\phi)^2\right) = \frac{1}{r^2}\left(1 + \left(\frac{\theta}{2}\right)^2\right) = \frac{1}{r^2}\left(1 + \frac{\theta^2}{4}\right)$$

$$= \frac{1}{r^2}\left(1 + (-\cot\theta/2)^2\right) = \frac{1}{r^2}\left[\operatorname{cosec}^2\theta/2\right] = \frac{1}{r^2}\,\frac{1}{\sin^2\theta/2}.$$

$\Rightarrow p^2 = r^2 \sin^2\theta/2 = r^2\left[\frac{1-\cos\theta}{2}\right]^2 = r^2\left(\frac{a}{r}\right)^2 = r^2\left[\frac{2a}{r}\right]$

$= \frac{r^2}{4}\, r^2\left(\frac{a}{r}\right) = ar$

$\Rightarrow p^2 = ar$

Q3 $a^2 = r^2\cos 2\theta$

Soln $2\log a = \log r^2 + \log\cos 2\theta = 2\log r + \log\cos 2\theta$

$\Rightarrow 0 = \frac{2}{r}r' + \frac{-\sin 2\theta\,(2)}{\cos 2\theta}$

$\Rightarrow \cot\phi = \sin 2\theta \tan 2\theta = \cot\left(\frac{\pi}{2} - 2\theta\right)$

$\Rightarrow \phi = \frac{\pi}{2} - 2\theta$

WKT, $\frac{1}{p^2} = \frac{1}{r^2}\left(1+(\cot\phi)^2\right) = \frac{1}{r^2}\left(1+\cot\left(\frac{\pi}{2}-2\theta\right)\right)$

$= \frac{1}{r^2}\left(\operatorname{cosec}^2\left(2\theta\frac{\pi}{2} - 2\theta\right)\right)$

$\Rightarrow p^2 = r^2\sin^2\left(\frac{\pi}{2}-2\theta\right) = r^2\left[1-\cos^2 2\theta\right]$

$\Rightarrow p^2 = r^2\left[\frac{a^2}{r^2}\right]^2 = r^2\frac{a^4}{r^4}$

$\Rightarrow P = \frac{a^2}{r}$

classmate
Date
Page

Q4 $a^m = r^m \cos m\theta$

Soln $m\log a = m\log r + \log\cos m\theta$

$\Rightarrow 0 = m\frac{1}{r}r' + \frac{-\sin m\theta\,(m)}{\cos m\theta}$

$\Rightarrow \cot\phi = \tan m\theta = \cot(\pi/2 - m\theta)$

$\frac{1}{p^2} = \frac{1}{r^2}(1+(\cot\phi)^2) = \frac{1}{r^2}(1+\tan^2 m\theta) = \frac{1}{r^2}\sec^2 m\theta$

$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\frac{1}{\cos^2 m\theta} \Rightarrow p^2 = r^2\cos^2 m\theta$

$\Rightarrow p^2 = r^2\left(\frac{a^m}{r^m}\right)^2 \Rightarrow p^2 = r^2\frac{a^{2m}}{r^{2m}} \Rightarrow p = \frac{r a^m}{r^m} = a^m r^{1-m}$

Q5 $r = ae^{m\theta}$

Soln $\log r = \log a + m\theta\log e$

$\Rightarrow \frac{1}{r}r' = 0 + \frac{m}{\theta}m$

$\Rightarrow \cot\phi = m$

WKT, $\frac{1}{p^2} = \frac{1}{r^2}(1+(\cot\phi)^2) = \frac{1}{r^2}(1+m^2)$

$\Rightarrow p^2 = \frac{r^2}{(1+m^2)} \Rightarrow p = \frac{r}{\sqrt{1+m^2}}$

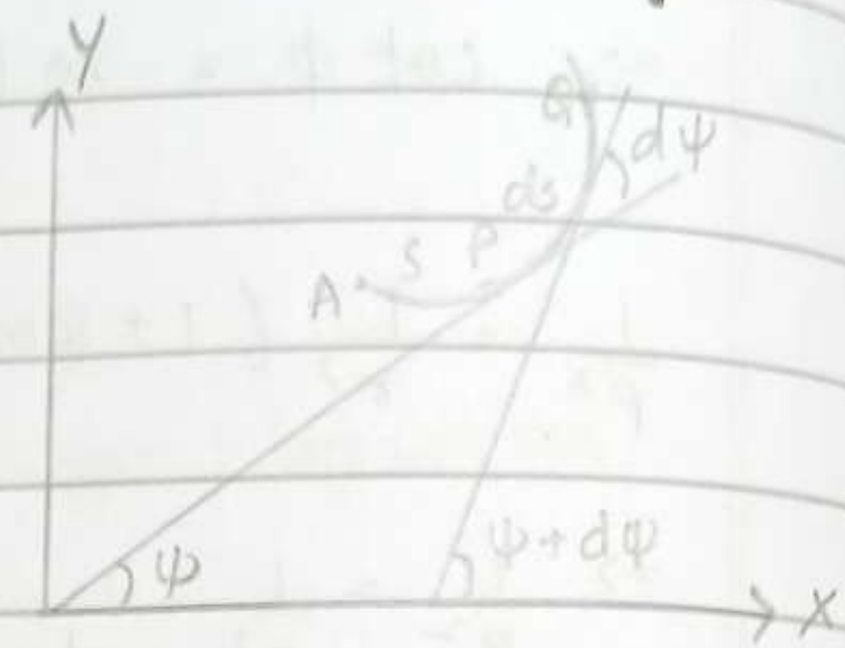
Not there

Curvature and radius of curvature

Let $y = f(x)$ be a cartesian curve and 's' is the actual length measured along the curve. Then the derivative of arc length can be evaluated by

$$\frac{ds}{dx} = \sqrt{1 + y_1^2}$$

where $y_1 = \frac{dy}{dx}$



Curvature :

Let P and Q be any two neighbouring points on the given curve. Let $\overset{\frown}{AP} = S$, $\overset{\frown}{PQ} = ds$, $\overset{\frown}{AQ} = S + ds$

The tangents at P and Q makes angle ψ and $(\psi + d\psi)$ with the x-axis so that the angle between b/n the tangents is $d\psi$ and is called 'total bending' or 'total curvature of the arc'. The average curvature of the arc PQ is $\frac{\delta\psi}{\delta s}$, and the limiting factor of the limiting value of the curvature $\lim_{\substack{Q \to P \\ \delta s \to 0}} \frac{\delta\psi}{\delta s} = \frac{d\psi}{ds} = k$ constant

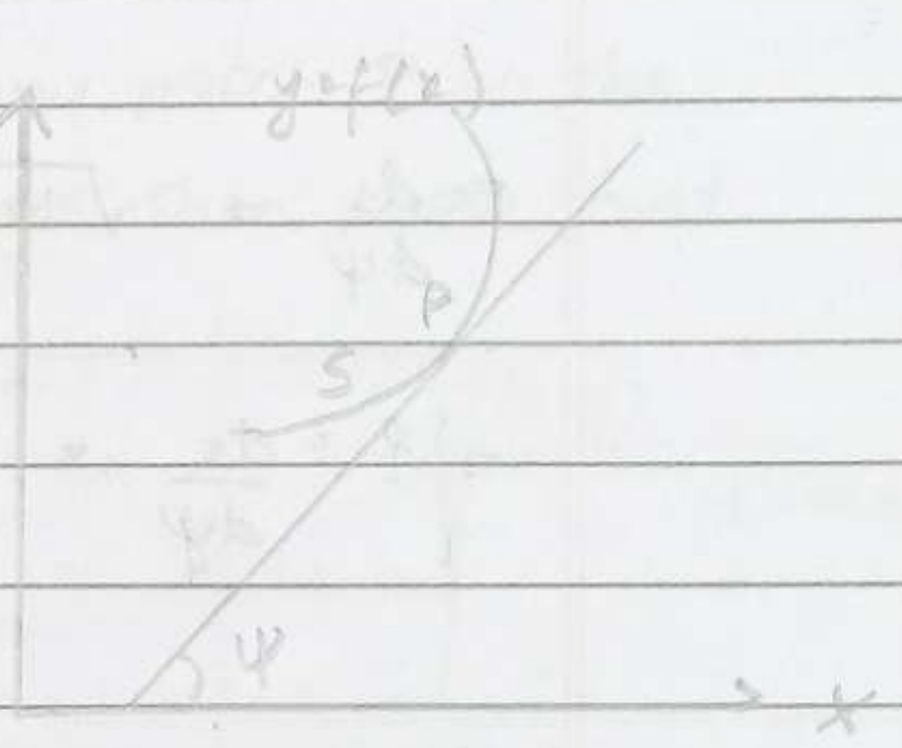
So that the curvature is the measurement of the rate of the bending of a curve at a point and given by, $k = \frac{d\psi}{ds}$

The radius of curvature is the reciprocal of the curvature and is given as $\rho = \frac{1}{k} = \frac{ds}{d\psi}$

Imp Derive the equation for Radius of curvature in a cartesian form. [for the curve $y = f(x)$]

$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$ where $y_1 = \frac{dy}{dx}$; $y_2 = \frac{dy_1}{dx}$

⇒ Let P be any point on the curve $y = f(x)$. Draw a tangent at P intersects x-axis at an angle ψ. Then, the slope of the tangent, $\tan\psi = \frac{dy}{dx}$



$\Rightarrow \tan\psi = \frac{dy}{dx} = y_1$

$\Rightarrow \psi = \tan^{-1}(y_1)$

diff w.r.t. x,

$\frac{d\psi}{dx} = \frac{1}{1+y_1^2}\left(\frac{dy_1}{dx}\right) = \frac{y_2}{1+y_1^2}$ —— (1)

W.K.T., $\rho = \frac{1}{k} = \frac{ds}{d\psi}$

Take, $\frac{ds}{d\psi} = \frac{ds}{dx} \cdot \frac{dx}{d\psi}$

But $\frac{ds}{dx} = \sqrt{1+y_1^2}$ (derivative of arc length)

From ①, $\frac{dy}{dx} = \frac{y_2}{1+y_1^2}$

$\Rightarrow \frac{dx}{d\psi} = \frac{1+y_1^2}{y_2}$

$\Rightarrow \frac{ds}{d\psi} = \sqrt{1+y_1^2}\left(\frac{1+y_1^2}{y_2}\right)$

$\Rightarrow \boxed{\rho = \frac{ds}{d\psi} = \frac{(1+y_1^2)^{3/2}}{y_2}}$

Note :

① The radius of curvature for a curve $r = f(\theta)$,

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$$

where $r_1 = dr/d\theta$

$r_2 = dr_1/d\theta = d^2r/d\theta^2$

② The radius of curvature for a parametric curve with $x = f(t)$ and $y = g(t)$.

$$\rho = \frac{(x_1^2 + y_1^2)^{3/2}}{x_1 y_2 - y_1 x_2}$$

where $x_1 = dx/dt$,

$x_2 = d^2x/dt^2$

$y_1 = dy/dt$

$y_2 = d^2y/dt^2$

Q1 If ρ be the radius of curvature at any point P on the parabola $y^2 = 4ax$ and 'S' be its focus. Then show that ρ^2 varies as $(SP)^3$.

Soln: $$\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$$

We have curve, $y^2 = 4ax$

$$2yy_1 = 4a$$

$$\Rightarrow y_1 = \frac{4a}{2y} = \frac{2a}{y}$$

$$\Rightarrow y_2 = -\frac{2a}{y^2} y_1$$

$$\Rightarrow y_2 = -\frac{2a}{y^2} \cdot \frac{2a}{y} = \frac{-4a^2}{y^3}$$

Now,

$$\rho = \frac{\left(1 + \left(\frac{2a}{y}\right)^2\right)^{3/2}}{\frac{-4a^2}{y^3}} = \frac{(1 + 4a^2/y^2)^{3/2}}{-4a^2/y^3} = \frac{\left(\frac{y^2 + 4a^2}{y^2}\right)^{3/2}}{-4a^2/y^3}$$

$$= \frac{\frac{(y^2+4a^2)^{3/2}}{(y^2)^{3/2}}}{-4a^2/y^3} = \frac{(y^2+4a^2)^{3/2}}{-4a^2} \quad [\because y^2 = 4ax]$$

$$\Rightarrow \rho = \frac{(4ax+4a^2)^{3/2}}{-4a^2} = \frac{(4a)^{3/2}(x+a)^{3/2}}{-4a^2}$$

$$\Rightarrow \rho^2 = \frac{(4a)^3(x+a)^3}{16a^4} \quad \text{——} (1)$$

Using distance formula,

$$\text{distance} = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$SP = \sqrt{(x-a)^2+(y-0)^2}$$

$$= \sqrt{x^2+a^2-2ax+y^2}$$

$$= \sqrt{x^2+a^2-2ax+4ax}$$

$$= \sqrt{(x+a)^2}$$

$$= x+a = SP$$

Substitute in (1),

$$\rho^2 = \frac{(4a)^3}{16 \cdot a^4}(SP)^3$$

$$= \frac{64a^3}{16a^4}(SP)^3$$

$$= \frac{4}{a}(SP)^3 \Rightarrow \rho^2 = (SP)^3$$

classmate
Date
Page

Q2 Prove that radius of curvature of a circle is constant.

Soln: For a circle, we have,

$$x^2+y^2=r^2$$

Let $r=a$,

$$x^2+y^2=a^2$$

$[x=r\cos\theta \text{ and } y=r\sin\theta] = [x=a\cos\theta \text{ and } y=a\sin\theta]$.

For parametric curve,

$$\rho = \frac{(x_1^2+y_1^2)^{3/2}}{x_1y_2-y_1x_2}$$

$$x_1=\frac{dx}{d\theta}=-a\sin\theta \qquad y_1=\frac{dy}{d\theta}=a\cos\theta$$

$$x_2=\frac{dx_1}{d\theta}=-a\cos\theta \qquad y_2=\frac{dy_1}{d\theta}=-a\sin\theta$$

$$\rho = \frac{((-a\sin\theta)^2+(a\cos\theta)^2)^{3/2}}{(-a\sin\theta)(-a\sin\theta)-(a\cos\theta)(-a\cos\theta)} = \frac{(a^2\sin^2\theta+a^2\cos^2\theta)^{3/2}}{a^2\sin^2\theta+a^2\cos^2\theta}$$

$$= \frac{(a^2)^{3/2}}{a^2} = \frac{a^3}{a^2} = a$$

$$\Rightarrow \rho \propto a$$

Q3 Show that the radius of curvature at any points of the cardiod. $r=a(1-\cos\theta)$ varies as $\sqrt{r}$.

Soln: $$\rho = \frac{(r^2+r_1^2)^{3/2}}{r^2+2r_1^2-rr_2}$$

$x_1 = a\sin\theta$

$y_2 = a\cos\theta$

$$\rho = \frac{[(a(1-\cos\theta))^2 + (a\sin\theta)^2]^{3/2}}{[a(1-\cos\theta)]^2 + 2a^2\sin^2\theta - a(1-\cos\theta)\cos\theta}$$

$$= \frac{[a^2(1+\cos^2\theta - 2\cos\theta) + a^2\sin^2\theta]^{3/2}}{a^2(1+\cos^2\theta - 2\cos\theta) + 2a^2\sin^2\theta - a^2\cos\theta + a^2\cos^2\theta}$$

$$= \frac{(a^2 + a^2\cos^2\theta - 2a^2\cos\theta + a^2\sin^2\theta]^{3/2}}{a^2 + a^2\cos^2\theta - 2a^2\cos\theta + 2a^2\sin^2\theta - a^2\cos\theta + a^2\cos^2\theta}$$

$$= \frac{[a^2 + a^2(\cos^2\theta + \sin^2\theta) - 2a^2\cos\theta]^{3/2}}{a^2 + 2a^2\cos^2\theta - 3a^2\cos\theta + 2a^2\sin^2\theta}$$

$$= \frac{(2a^2 - 2a^2\cos\theta)^{3/2}}{a^2 + 2a^2(\cos^2\theta + \sin^2\theta)}$$

$$= \frac{[2a^2(1-\cos\theta)]^{3/2}}{3a^2 - 3a^2\sin^2\theta}$$

$$= \frac{2^{3/2}\, a^3 (1-\cos\theta)^{3/2}}{3a^2(1-\cos\theta)}$$

$$= \frac{2\sqrt{2}}{3} a \sqrt{1-\cos\theta}$$

$$= \frac{2\sqrt{2}}{3} \sqrt{a}\, \sqrt{a(1-\cos\theta)}$$

$\Rightarrow r = a(1-\cos\theta)$

$\Rightarrow \sqrt{r} = \sqrt{a(1-\cos\theta)}$

$\rho = \frac{2\sqrt{2}}{3} \sqrt{a}\, \sqrt{r}$ $\Rightarrow \rho \propto \sqrt{r}$

Q1 $x = a\cos^3\theta$; $y = a\sin^3\theta$; find r.c. at any point on thes

Soln: $S = \dfrac{(x_1^2 + y_1^2)^{3/2}}{x_1 y_2 - y_1 x_2}$ points

$x_1 = a\,3\sin^2\theta\, 3a\cos^2\theta(-\sin\theta)$

$x_2 = -3a[2\cos\theta(-\sin\theta) + \cos^3\theta\cos\theta] = -3a[-2\cos^2\theta\sin^2\theta + \cos^2\theta\cos\theta]$

$= -3a\cos^3\theta + 6a\sin^2\theta\cos\theta$

$y_1 = a\,3\sin^2\theta\cos\theta$

$y_2 = 3a(\sin^3\theta) + 6a\cos^2\theta\sin\theta$

$S = \dfrac{[(-3a\cos^2\theta\sin\theta)^2 + (a3\sin^2\theta\cos\theta)^2]^{3/2}}{(-3a\cos^2\theta\sin\theta)(-3a\sin^3\theta + 6a\cos^2\theta\sin\theta) - [3a\sin^2\theta\cos\theta(-3a\cos^3\theta + 6a$

$= \dfrac{[9a^2\cos^4\theta\sin^2\theta + 9a\sin^4\theta\cos^2\theta]^{3/2}}{9a^2\cos^2\theta\sin^4\theta - 18a^2\cos^4\theta\sin^2\theta + 9a^2\cos^4\theta\sin^2\theta - 18a^2\sin^4\theta\cos^2\theta}$

$= \dfrac{[9a^2\cos^2\theta\sin^2\theta(\cos^2\theta + \sin^2\theta)]^{3/2}}{-9a^2\cos^2\theta\sin^4\theta - 9a^2\cos^4\theta\sin^2\theta}$

$= \dfrac{(3a\cos\theta\sin\theta)^{2\times 3/2}}{-9a^2\cos^2\theta\sin^2\theta} = \dfrac{(3a\cos\theta\sin\theta)^3}{-(3a\cos\theta\sin\theta)^2} = -3a\cos\theta\sin\theta$

$\Rightarrow S = |-3a\cos\theta\sin\theta|$

Find the radius of curvature at any point on a cycloid, $x = a(\theta - \sin\theta)$, $y = a(1-\cos\theta)$

Soln: $$S = \frac{(x_1^2 + y_1^2)^{3/2}}{x_1 y_2 - x_2 y_1}$$

$x = a(\theta - \sin\theta)$ $\qquad$ $y = a(1-\cos\theta)$

$x_1 = a(1-\cos\theta)$ $\qquad$ $y_1 = a(\ \ \sin\theta)$

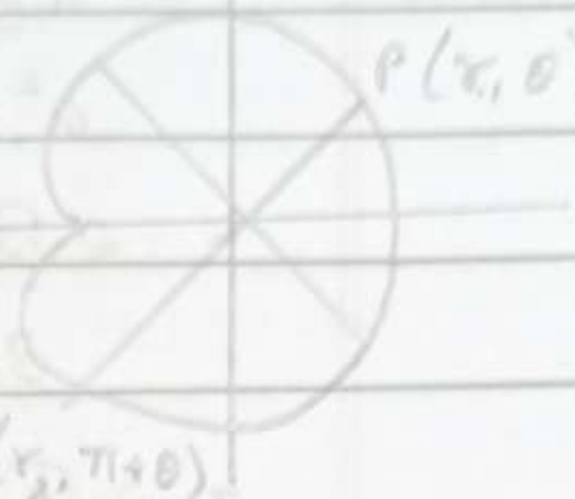
$x_2 = a(\sin\theta)$ $\qquad$ $y_2 = a(\cos\theta)$

$$S = \frac{[a^2(1-\cos\theta)^2 + a^2(\ \ \sin\theta)^2]^{3/2}}{a(1-\cos\theta)a(\cos\theta) - a\sin\theta a(\ \ \sin\theta)}$$

$$= \frac{[a^2(1+\cos^2\theta - 2\cos\theta) + a^2\ \ \sin^2\theta\ \ \ \ \ \ \ \ \ \ )]^{3/2}}{a^2(\cos\theta - \cos^2\theta) - a^2(\ \ \ \ \ \ \sin^2\theta)}$$

$$= \frac{[a^2 + a^2\cos^2\theta - 2a^2\cos\theta \ \ \ \ a^2\sin^2\theta \ \ \ \ \ \ \ \ \

If ρ_1 and ρ_2 are the R.C. at the extreme point of chord, which passes through pole of cardiod $r = a(1+\cos\theta)$, proove that $\rho_1^2 + \rho_2^2 = \frac{16a^2}{9}$



Soln: Consider a chord PQ which passes through the pole. Then for a polar curve, we have the R.C.,

$$\rho = \frac{(r^2 + r_1'^2)^{3/2}}{r^2 + 2r_1'^2 - r r_1''}$$

Let $r_1 = a(1+\cos\theta)$ and $r_2 = a(1+\cos(\pi+\theta))$ be the curves at point P and Q. ~~Then, we have the respective R.C.,~~

For a curve $r_1 = a(1+\cos\theta)$,

$$\rho_1 = \frac{(r^2 + r_1'^2)^{3/2}}{r^2 + 2r_1'^2 - r r_1''}$$

$r_1' = -a\sin\theta$

$r_1'' = -a\cos\theta$

$$\rho_1 = \frac{(a^2(1+\cos\theta)^2 + a^2\sin^2\theta)^{3/2}}{a^2(1+\cos\theta)^2 + 2a^2\sin^2\theta - a(1+\cos\theta)(-a\cos\theta)}$$

$$= \frac{[a^2 + a(1+\cos^2\theta + 2\cos\theta) + a^2\sin^2\theta]^{3/2}}{a^2(1+\cos^2\theta + 2\cos\theta) + 2a^2\sin^2\theta + a(a\cos\theta + a\cos^2\theta)}$$

$$= \frac{[a^2 + a^2\cos^2\theta + 2a^2\cos\theta + a^2\sin^2\theta]^{3/2}}{a^2 + a^2\cos^2\theta + 2a^2\cos\theta + 2a^2\sin^2\theta + a^2\cos\theta + a^2\cos^2\theta}$$

$$= \frac{[a^2 + a^2 + 2a^2\cos\theta]^{3/2}}{a^2 + 3a^2\cos\theta + 2a^2\sin^2\theta + 2a^2\cos^2\theta}$$

$$= \frac{[2a^2 + 2a^2\cos\theta]^{3/2}}{a^2 + 2a^2 + 3a^2\cos\theta} = \frac{[2a^2(1+\cos\theta)]^{3/2}}{3a^2(1+\cos\theta)}$$

$$= \frac{2\sqrt{2}\,a^3\,(1+\cos\theta)^{3/2}}{3a^2\,(1+\cos\theta)} = \frac{2\sqrt{2}}{3}\,a\,\sqrt{1+\cos\theta}$$

$$= \frac{2\sqrt{2}}{3}\,a\,\frac{\sqrt{r}}{\sqrt{a}}$$

$$= \frac{2\sqrt{2}}{3}\sqrt{ra}$$

$$=$$

$$\Rightarrow \rho^2 = \frac{8}{9}ar = \frac{8}{a}a(a)(1+\cos\theta) = \frac{8}{9}a^2(1+\cos\theta)$$

$$=$$

At θ, $\rho_2^2 = \frac{8a^2}{9}(1-\cos\theta)$.

$$\rho_1^2 + \rho_2^2 = \frac{8a^2}{9}[1+\cos\theta+1-\cos\theta] = \frac{16a^2}{9}$$

Show that R.C. at $x = \pi/2$, $y = 4\sin x - \sin 2x$ is $\frac{5\sqrt{5}}{4}$

Soln: $\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$

$y = 4\sin x - \sin 2x$

$\Rightarrow \frac{dy}{dx} = 4\cos x - 2\cos 2x = 4\cos x - 2\cos 2x$

$y_2 = -4\sin x + 2\cos 2x = 2\cos 2x - 4\sin x.$

$$① \Rightarrow \left.\frac{dy}{dx}\right|_{x=\pi/2} = y_1 = 1\cos\frac{\pi}{2} - 2\cos 2\frac{\pi}{2} = 0 - 2(-1) = 2$$

$$② \Rightarrow \left.\frac{d^2y}{dx^2}\right|_{x=\pi/2} = y_2 = 1\cos 2\frac{\pi}{2} - 1\sin\frac{\pi}{2} = 1(-1) - 1(0) = -1$$

$$\Rightarrow \rho = \frac{(1+2^2)^{3/2}}{|-1|} = \frac{(5)^{3/2}}{4} = \frac{5\sqrt{5}}{4}$$

Find the radius of curvature of the curve $r^n = a^n \cos n\theta$

Soln: $$\rho = \frac{(r^2+r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$$

$$n\log r = n\log a + \log\cos n\theta$$

$$\Rightarrow \frac{n}{r} r' = \frac{n}{a}(0) + \frac{(-\sin\theta)n}{\cos n\theta}$$

$$\Rightarrow r' = \frac{dr}{d\theta} = -r\tan n\theta$$

$$\Rightarrow r_2 = -[rn\sec^2\theta + \tan n\theta\, r_1]$$

$$= -rn\sec^2\theta + r\tan^2 n\theta$$

$$\rho = \frac{(r^2 + (-r\tan n\theta)^2)^{3/2}}{r^2 + 2(-r\tan n\theta)^2 - r(-rn\sec^2 n\theta + r\tan^2 n\theta)}$$

$$= \frac{(r^2 + r^2\tan^2 n\theta)^{3/2}}{r^2 + 2r^2\tan^2 n\theta + r^2 n\sec^2 n\theta - r^2\tan^2 n\theta}$$

$$= \frac{(r^2\sec^2\theta)^{3/2}}{r^2(1+\tan^2 n\theta) + r^2 n\sec^2 n\theta}$$

$= \frac{(r\sec n\theta)^3}{r^3\sec^3 n\theta \, r r^2 n \sec^2 n\theta}$

$= \frac{r\sec n\theta}{1+n} = \frac{r}{1+n}\left(\frac{1}{\cos n\theta}\right)$

$= \frac{r}{(1+n)} \frac{a^n}{r^n}$

$\rho = \frac{a^n}{(1+n) r^{n-1}}$

about the point $x=a$ will be a series in powers of $x-a$ known as Taylor's series.

$$y = f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''a + \frac{(x-a)^4}{4!} f''''$$

.....

If $a=0$, then,

$$y = f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \cdots$$

This is M series.

Obtain taylor series expansion of $\sin x$ at $x = \pi/2$ upto the terms containg $(x - \pi/2)^4$.

Soln: $y = f(x) = \sin x$

$f(x) = f(a) = \sin(\pi/2) = 1$

$f'(x) = \cos x$

$f'(a) = \cos(\pi/2) = 0$

$f''(x) = -\sin x$

$f''(a) = -\sin \pi/2 = -1$

$f'''(x) = -\cos x$

$f'''(a) = -\cos(\pi/2) = 0$

$f^{iv}(x) = \sin x$

$f^{iv}(a) = \sin \pi/2 = 1$

$$y = f(x) = \sin x = 1 + \frac{f(x-\pi/2)(0)}{1!} + \frac{(x-\pi/2)^2(-1)}{2!} + \frac{(x-\pi/2)^2(0)}{3!} + \frac{(x-\pi/2)^4(1)}{4!}$$

$$\Rightarrow \sin x = 1 - \frac{(x-\pi/2)^2}{2} + \frac{(x-\pi/2)^4}{24}$$

Q2 Of $\log x$ at $x = 1$ upto $(x-\pi/2)^4$

Soln: $y = f(x) = \log x$

$f(x) = f(a) = \log a = \log 1 = 0$

$\Rightarrow f(a) = \log a = \log 1 = 0$

$\Rightarrow f'(x) = 1/x$

$\Rightarrow f'(a) = 1/a = 1/1 = 1$

$f''(x) = -1/x^2$

$f''(a) = -1/a^2 = -1/1 = -1$.

$\Rightarrow f'''(x) = 2/x^3$

$\Rightarrow f'''(a) = 2/a^3 = 2/1 = 2$

$\Rightarrow f^{iv}(x) = -6/x^4$

$\Rightarrow f^{iv}(a) = -6/a^4 = -6$.

$$y = f(x) = 0 + (x-1)(1) + \frac{(x-1)^2}{2!}(-1) + \frac{(x-1)^3}{3!}(2) + \frac{(x-1)^4}{4!}(-6)$$

$$\Rightarrow y = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{6}(2) - \frac{6}{24}(x-1)^4 + \dots\dots$$

Q3. of $\cos x$ at $x = \pi/4$.

Soln: $f'(x) = -\sin x$ $\quad f''(x) = -\cos x$ $\quad f'''(x) = \sin x$

$f'(a) = \frac{-1}{\sqrt{2}}$ $\quad f''(a) = \frac{-1}{\sqrt{2}}$ $\quad f'''(a) = \frac{1}{\sqrt{2}}$

$f^{iv}(x) = \cos x$

$f^{iv}(a) = \frac{1}{\sqrt{2}}$

$$y = f(x) = \frac{-1}{\sqrt{2}} + \frac{(x-\pi/4)}{1!}\left(\frac{-1}{\sqrt{2}}\right) + \frac{(x-\pi/4)^2}{2!}\left(\frac{-1}{\sqrt{2}}\right) + \frac{(x-\pi/4)^3}{3!}\left(\frac{1}{\sqrt{2}}\right)$$

$$+ \frac{(x-\pi/4)^4}{4!}\left(\frac{1}{\sqrt{2}}\right)$$

$$\Rightarrow \cos x = \frac{-1}{\sqrt{2}} - \frac{(x-\pi/4)}{\sqrt{2}} - \frac{(x-\pi/4)^2}{2\sqrt{2}} + \frac{(x-\pi/4)^3}{6\sqrt{2}} + \frac{(x-\pi/4)^4}{24\sqrt{2}}$$

classmate
Date
Page

Q4. Find the series of the function $y = e^x$.

Soln: $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

$f'(x) = e^x \Rightarrow f'(0) = 1$

$f''(x) = e^x \Rightarrow f''(0) = 1$

$f'''(x) = e^x \Rightarrow f'''(0) = 1$

$f^{iv}(x) = e^x \Rightarrow f^{iv}(0) = 1.$

$\Rightarrow f(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$

Q5. Find the series of the function $y = \sin x$.

Soln $f'(x) = \cos x \Rightarrow f'(0) = \cos 0 = \pi/2\ 1$

$f''(x) = -\sin x \Rightarrow f''(0) = 0 - \sin 0 = \pi\ 0$

$f'''(x) = -\cos x \Rightarrow f'''(0) = -1 - \cos 0 = -\pi/2\ -1$

$f^{iv}(x) = \sin x \Rightarrow f^{iv}(0) = 0 - \sin 0 = \pi\ 0$

$f(x) = \frac{\pi}{2} + x\ 1 + x(0) + \frac{x^2}{2}(0) + \frac{x^3}{6}(-1) + \frac{x^4}{24}(0)$

$= 1 + \frac{x^2}{2} - \frac{x^3}{6} + \dots$

Q6 $y = \log(1+x)$ at $x=0$,

Soln: $f'(x) = \frac{1}{1+x} \Rightarrow f'(0) = 1$; $f(0) = \log 1 = 0$

$f''(x) = \frac{-1}{(1+x)^2} \Rightarrow f''(0) = -1$

$f'''(x) = +\frac{2}{(1+x)^3} \Rightarrow f'''(0) = 2$

$f^{iv}(x) = -\frac{2(1)}{(1+x)^8} = \frac{-8}{(1+x)^8} \Rightarrow f^{iv}(0) = -8$

$$f(x) = 0 + x(1) + \frac{x^2}{2}(-1) + \frac{x^3}{6}(2) + \frac{x^4}{24}(-8)$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{3},$$

Q7 $y = (1+x)^m$

$f'(x) = m(1+x)^{m-1}$

$f''(x) = m(m-1)(1+x)^{m-2} = (m^2-m)(1+x)^{m-2}$

$f'''(x) = m(m-1)(m-2)(1+x)^{m-3}$

$$(1+x)^m = 1 + \frac{x}{1!}m + \frac{x^2}{2!}m(m-1) + \frac{x^3}{3!}(m)(m-1)(m-2) + \dots$$

H/W
- $y = \tan x$
- $y = e^{-x}$
- $y = \cos x$

2 descriptive
5-10 MCQ's

classmate
Date
Page

$y = \tan x$

$y\ f(x) = \tan x$

$f(0) = 0$

$f'(x) = \sec^2 x$

$f'(0) = \sec^2(0) = 1$

$f''(x) = 2\sec x \tan x \sec x$

$f''(0) = 2\sec(0)\tan(0)\sec(0) = 0$

$f'''(0) = 2\sec^2(0)(2\tan(0) + \sec 0$

$= 2$

$f(x) = 0 + \frac{x(1)}{1!} + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(2) + \cdots$

$= x + \frac{2x^3}{6}$

$y = e^{-x}$

$f(x) = e^{-x}$

$f(0) = 1$

$f'(x) = -e^{-x}$ $\quad f'(0) = -e^0 = -1$

$f''(x) = e^{-x}$ $\quad f''(0) = e^0 = 1$

$f'''(x) = -e^{-x}$ $\quad f'''(0) = -1$

$f^{iv}(x) = e^{-x}$ $\quad f^{iv}(0) = 1$

$f(x) = 1 + \frac{x}{1!}(-1) + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(-1) + \frac{x^4}{4!}(1) + \cdots$

$$= 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$y = \cos x$

$f(x) = \cos x$

$f(0) = 1$

$f'(x) = -\sin x$ $\quad f'(0) = -\sin 0$

$f''(x) = -\cos x$ $\quad f''(0) = -1$

$f'''(x) = \sin x$ $\quad f'''(0) = 0$

$f^{iv}(x) = \cos x$ $\quad f^{iv}(0) = -1$

$$f(x) = 1 + \frac{x}{1!}(0) + \frac{x}{2!}(-1) + \frac{x}{3!}(0) + \frac{x}{4!}(-1) + \cdots$$

$$= 1 - \frac{x}{2} + \frac{x}{24} + \cdots$$

Radius of curvature in Pedal Form

The radius of curvature for pedal curve is given by,

$$\rho = r \cdot \frac{dr}{dp}$$

Find the radius of curvature for the curve, $Pa^2 = r^3$.

Soln: diff. curve w.r.t. P,

$$a^2 = 3r^2 \frac{dr}{dP}$$

$$\Rightarrow \frac{dr}{dP} = \frac{a^2}{3r^2}$$

$$\Rightarrow \rho = r \cdot \frac{dr}{dP} = \frac{r a^2}{3r^2} = \frac{a^2}{3r}$$

Show that for the ellipse in the Pedal form,

$\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} - \frac{r^2}{a^2 b^2}$. The r.c. at a point (p, r) is

$\frac{a^2 b^2}{p^3}$

Soln: $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} - \frac{r^2}{a^2 b^2}$

diff. w.r.t. P,

$$\Rightarrow \frac{-2}{p^3} = 0 + 0 - \frac{2r\, dr/dP}{a^2 b^2}$$

$$\Rightarrow \frac{1}{p^3} = \frac{r}{a^2 b^2} \frac{dr}{dP}$$

$$\Rightarrow \frac{dr}{dP} = \frac{a^2 b^2}{p^3 r}$$

$$\Rightarrow \rho = r \frac{dr}{dP}$$

$$\Rightarrow \rho = r \frac{a^2 b^2}{p^3 r}$$

$$\Rightarrow \rho = \frac{a^2 b^2}{p^3}$$

Show that p.c. of $r\cos 2\theta = a$ is $\left|\frac{r^4}{3p^3}\right|$.

Soln: $r\cos 2\theta = a$

diff. w.r.t. θ,

$$\frac{dr}{d\theta}\cos 2\theta + 2r(-\sin 2\theta) = 0$$

$$\Rightarrow \frac{dr}{d\theta}\cos 2\theta = 2r\sin 2\theta$$

$$\Rightarrow \frac{dr}{d\theta} = 2r\tan 2\theta$$

$$=$$

Pedal eqn., $\frac{1}{p^2} = \frac{1}{r^2}\left(1 + \frac{1}{r^2}\left(\frac{dr}{d\theta}\right)^2\right)$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\left[1 + \frac{1}{r^2}4r^2\tan^2 2\theta\right]$$

$$= \frac{1}{r^2} + \frac{4}{r^4}r^2\tan^2 2\theta$$

$$= \frac{1}{r^2} + \frac{4}{r^2}\tan^2 2\theta \qquad = \frac{1}{r^2}(1 + 4\tan^2 2\theta)$$

$$= \frac{1}{r^2}\left[1 + 4(-1 + \sec^2 2\theta)\right] = \frac{1}{r^2}\left[1 + 4(\sec^2 2\theta - 1)\right]$$

From curve,

$$\frac{r}{a} = \sec 2\theta$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{r^2}\left[1 + 4\left(\frac{r^2}{a^2} - 1\right)\right].$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{\gamma^2}\left[1 + \frac{4\gamma^2}{a^2} - 1\right] = \frac{1}{\gamma^2}\left[\frac{4\gamma^2}{a^2} - 3\right]$$

$$\frac{1}{p^2} = \frac{4}{a^2} - \frac{3}{\gamma^2}$$

diff. w.r.t. P,

$$-\frac{2}{p^3} = 0 + \frac{3(2)}{\gamma^3}\frac{d\gamma}{dP}$$

$$\Rightarrow \frac{d\gamma}{dP} = \frac{-2}{p^3}\frac{\gamma^3}{6} = \frac{-\gamma^3}{p^3(3)}$$

$$\rightarrow f = \frac{\gamma}{\gamma}\frac{d\gamma}{dP} = \frac{\gamma}{\gamma}\left(\frac{-\gamma^3}{p^3(3)}\right) = \frac{-\gamma^4}{3p^3}$$

Partial Differentiation

A partial derivative is the derivative of multi-variable function w.r.t. one independent variable, treating all other independent variables as constant.

Let $f(x, y, z, \ldots)$ is a multi-variable function, then the partial derivative w.r.t. 'x' is defined as,

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y, z, \ldots) - f(x, y, z, \ldots)}{h}$$

Here, $(y, z, \ldots)$ are treated as constants while diff. w.r.t. x.

The notation used for partial differentiation is, '∂ (doh, dow)'.

Example: $u = 3x^2y + 6xy^2 + z$ 7

$$\Rightarrow \frac{du}{\partial x} = 6xy + 6y^2 + 0$$

$$\Rightarrow \frac{\partial u}{\partial y} = 3x^2 + 6x(2y) = 3x^2 + 12xy.$$

Order of partial derivatives

Let $u = f(x, y)$ be a function. Then,

$u_x = \frac{\partial u}{\partial x}$, $u_y = \frac{\partial u}{\partial y}$ are first order.

$u_{xx} = \frac{\partial^2 u}{\partial x^2}$, $u_{yy} = \frac{\partial^2 u}{\partial y^2}$ are second order

$u_{xy} = \frac{\partial^2 u}{\partial x \partial y}$ is also second order. } Mixed order.

$u_{yx} = \frac{\partial^2 u}{\partial y \partial x}$ is also second order }

$u_{xy} = u_{yx}$.

Q2. Find u_x, u_y, u_{xx}, u_{yy}, u_{xy}, u_{yx}.

(i) $u = xy^2 + x^2y$

Soln: $u_x = \frac{\partial u}{\partial x} = y^2 + 2xy$ $\quad u_{yy} = \frac{\partial^2 u}{\partial y^2} = 2x + 0$

$u_y = \frac{\partial u}{\partial y} = 2xy + x^2$ $\quad u_{xy} = \frac{\partial^2 u}{\partial x \partial y} = 2y + 2x$

$u_{xx} = \frac{\partial^2 u}{\partial x^2} = y^2 + 2y$ $\quad u_{yx} = u_{xy} = 2y + 2x$

(i) $z = x^2 \sin y - y^2 \sin x$

$\Rightarrow z_x = 2x \sin y - y^2 \cos x$

$z_y = x^2 \cos y - 2y \sin x$

$z_{xx} = 2 \cdot \sin y + 2y^2 \sin x$

$z_{yy} = -x^2 \sin y - 2 \sin x$

$z_{xy} = +2x \cos y - 2y \cos x = z_{yx}$

(ii) $z = x^3 + y^3 - 3axy;$

z_x, z_y, z_{xx}

$\Rightarrow z_x = 3x^2 + y^3 - 3ay$

$z_y = x^3 + 3y^2 - 3ax$

$z_{xx} = 6x$

$z_{yy} = 6y$

$z_{xy} = -3a = z_{yx}$

(iii) $z = \tan^{-1}\left(\frac{y}{x}\right)$

$\Rightarrow z_x = \frac{1}{1+\left(\frac{y}{x}\right)^2} y\left(-\frac{1}{x^2}\right) = \frac{x^2}{x^2+y^2}\left(\frac{-y}{x^2}\right) = \frac{-y}{x^2+y^2} = z_{y^1}$

$z_y = \frac{1}{1+\left(\frac{y}{x}\right)^2}\left(\frac{1}{x}\right) = \frac{x^2}{x^2+y^2}\,\overline{x} = \frac{x}{x^2+y^2}$

$z_{xx} = -y\left(\frac{1}{x^2+y^2}\right) = +y\,\frac{(2x)}{(x^2+y^2)^2}$

$$z_{yy} = -\frac{x(2y)}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$$

$$z_{xy} = -\frac{[1(x^2+y^2) - y(2y)]}{(x^2+y^2)^2} = \frac{-x^2+y^2+2y^2}{(x^2+y^2)^2} = z_{yx}$$

If $u = \frac{e^{x+y}}{e^x+e^y}$. PT.. $u_x + u_y = u$

Soln: $u_x \Rightarrow \frac{e^x \cdot e^y}{e^x+e^y} = e^x e^y \cdot \frac{(e^x e^y + 0)(e^x+e^y) - (e^x e^y)(e^x)}{(e^x+e^y)^2}$

$$= \frac{(e^x e^y)(e^x+e^y) - e^x e^x e^y}{(e^x+e^y)^2}$$

$$= \frac{e^x e^x e^y + e^y e^x e^y - e^x e^x e^y}{(e^x+e^y)^2}$$

$$= \frac{e^{2y} e^x}{(e^x+e^y)^2}$$

$$u_y = \frac{(e^x e^y)(e^x+e^y) - e^x e^y (e^y)}{(e^x+e^y)^2} = \frac{e^x e^y e^x + e^x e^y e^y - e^y e^x e^y}{(e^x+e^y)^2}$$

$$= \frac{e^{2x} e^y}{(e^x+e^y)^2}$$

$$u_x + u_y = \frac{e^{2y} e^x}{(e^x+e^y)^2} + \frac{e^{2x} e^y}{(e^x+e^y)^2} = \frac{e^x e^y (e^y + e^x)}{(e^x+e^y)^2}$$

$$= \frac{e^x e^y e^{x+y}}{(e^x+e^y)^2}$$

$u = \log(x^2+y^2)$, p.t. $u_{xy} = u_{yx}$

$$\rightarrow u_x = \frac{1}{x^2+y^2} 2x(2) = \frac{4x}{x^2+y^2}$$

$$u_y = \frac{4y}{x^2+y^2}$$

$$u_{xy} = 4\left[\frac{(1)(x^2+y^2)-2xy}{(x^2+y^2)^2}\right] = \frac{4(x^2+y^2-2xy)}{(x^2+y^2)^2} = \frac{4(x-y)^2}{(x^2+y^2)^2}$$

$$u_{yx} = 4\left[\frac{(1)(x^2+y^2)-2xy}{(x^2+y^2)^2}\right]$$

$$= 4\left[\frac{x^2+y^2-2xy}{(x^2+y^2)^2}\right]$$

$\therefore u_{xy} = u_{yx}$

If $x = f(x+ct) + \phi(x-ct)$ p.t. $u_{tt} = c^2 u_{xx}$.

Soln: $x =$

$u = f(v) + \phi(w)$; $v = x+ct$; $w = x-ct$

$$\frac{du}{dt} = \frac{\partial f}{\partial v}\frac{dv}{dt} + \frac{\partial \phi}{\partial w}\frac{dw}{dt}$$

$$\frac{\partial v}{\partial t} = c \quad ; \quad \frac{\partial w}{\partial t} = -c$$

$$\frac{\partial u}{\partial t} = f'(v)c + \phi'(w)(-c) = cf'(x+ct) - c\phi'(x-ct)$$

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial \phi}{\partial w}\frac{\partial w}{\partial x}$$

$\left(\frac{\partial v}{\partial x} = 1 \quad \& \quad \frac{\partial w}{\partial x} = 1\right)$

$$\Rightarrow \frac{\partial u}{\partial x} = f'(v)(1) + \phi'(w)(1) = f'(x+ct) + \phi'(x-ct)$$

$$\Rightarrow \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\left(cf'(x+ct) - c\phi'(x-ct)\right)$$

$$\Rightarrow \frac{\partial^2 u}{\partial t^2} = c \cdot f''(x+ct) \cdot c - c\phi''(x-ct)(-c)$$

$$= c^2 f''(x+ct) + \phi''(x-ct)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(f'(x+ct) + \phi'(x-ct)\right) = f''(x+ct) + \phi''(x-ct)$$

$$\Rightarrow u_{tt} = \frac{\partial^2 u}{\partial t^2} = c^2\left(f''(x+ct) + \phi''(x-ct)\right)$$

$$\Rightarrow u_{xx} = \frac{\partial^2 u}{\partial x^2} = f''(x+ct) + \phi''(x-ct)$$

$$\Rightarrow u_{tt} = c^2 u_{xx}$$

Homogenous Function

A function $u = f(x,y)$ is said to be a homogeneous of degree 'n', if it can be expressed in the form of.

$$x^n\left[g\left(\frac{y}{x}\right)\right] \quad \text{or} \quad y^n\left[g\left(\frac{x}{y}\right)\right]$$

where g is any arbitrary function.

Ex: $u = 3x + 4y$

$u = x\left(3 + 4\frac{y}{x}\right)$

It is a homogenous eqn of degree 1.

Q1 $u = x^2y + y^2x$

Soln: $u = x^3\left[\frac{y}{x} + \frac{y^2}{x^2}\right]$

It is homogenous of degree 3.

Euler's Theorem

If $u = f(x, y)$ a homogenous fnc. of degree n. In x and y. Then prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$

and hence deduce that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 y}{\partial x \partial y} + y^2\frac{\partial^2 y}{\partial y^2}$

$= n(n-1)u.$

→ If $u = f(x, y)$ a homogenous fnc.,

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u$$

$u = x^n g\left(\frac{y}{x}\right)$ —— ①

diff ① w.r.t. x,

$\frac{\partial u}{\partial x} = x^n g'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right) + g\left(\frac{y}{x}\right) n x^{n-1}$

$$= -yx^{n-2} g'\left(\frac{y}{x}\right) + nx^{n-1} g\left(\frac{y}{x}\right) \quad \text{—— (2)}$$

$$\therefore x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu \quad \text{—— (3)}$$

diff. w.r.t. y,

$$\frac{\partial u}{\partial y} = x^n g'\left(\frac{y}{x}\right)\frac{1}{x}(1) = x^{n-1} g'\left(\frac{y}{x}\right)$$

$$\text{LHS} \Rightarrow x\frac{\partial y}{\partial x} + y\frac{\partial u}{\partial y}$$

$$\Rightarrow x\left[-yx^{n-2} g'\left(\frac{y}{x}\right) + nx^{n-1} g\left(\frac{y}{x}\right)\right] + y\left[x^{n-1} g'\left(\frac{y}{x}\right)\right]$$

$$= -x^{n-1} y\, g'\left(\frac{y}{x}\right) + nx^n g\left(\frac{y}{x}\right) + yx^{n-1} g'\left(\frac{y}{x}\right)$$

$$= nx^n g\left(\frac{y}{x}\right)$$

diff (3), w.r.t. x,

$$x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x \partial y} = n\frac{\partial u}{\partial x} \quad \text{—— (4)}$$

multiply (4) by x,

$$x^2\frac{\partial^2 u}{\partial x^2} + x\frac{\partial y}{\partial x} + xy\frac{\partial^2 u}{\partial x \partial y} = nx\frac{\partial u}{\partial x} \quad \text{—— (5)}$$

diff (3) w.r.t. y,

$$x\frac{\partial^2 y}{\partial y \partial x} + y\frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} = n\frac{\partial u}{\partial y} \quad \text{—— (6)}$$

multiply (6) by y,

$$xy\frac{\partial^2 y}{\partial y \partial x} + y^2\frac{\partial^2 u}{\partial y^2} + y\frac{\partial u}{\partial y} = ny\frac{\partial u}{\partial y} \quad \text{—— (7)}$$

eqn ③ + ④,

$$\left[x^2\frac{\partial^2 u}{\partial x^2} + x\frac{\partial u}{\partial x} + xy\frac{\partial^2 u}{\partial x \partial y}\right] + \left[xy\frac{\partial^2 y}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} + y\frac{\partial y}{\partial y}\right] = nx\frac{\partial u}{\partial x} + ny\frac{\partial u}{\partial y}$$

$$\Rightarrow x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)$$

$\Rightarrow$ from ③,

$$x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + nu = n^2 u$$

$$\Rightarrow x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} = n(n-1)u$$

Q1 $u = \log\left(\frac{x^4+y^4}{x+y}\right)$ P.T., $x\frac{du}{dx} + y\frac{du}{dy} = 3$.

Soln: $u = \log\left(\frac{x^4+y^4}{x+y}\right)$

$\Rightarrow e^u = e^{\log\left(\frac{x^4+y^4}{x+y}\right)}$

$\Rightarrow e^u = \frac{x^4+y^4}{x+y} = \frac{x^4}{x}\frac{(1+y^4/x^4)}{(1+y/x)} = x^3\left[\frac{1+(y^4/x^4)}{1+(y/x)}\right]$

This is a homogenous fnc. in x.

By. euler's theorem.

$$\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

$$\Rightarrow xe^u\frac{\partial u}{\partial x} + ye^u\frac{\partial u}{\partial y} = 3e^u$$

$$\Rightarrow e^u\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) = 3e^u$$

$$\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$$

Q2. If $u = \sin^{-1}\left(\frac{x^2y^2}{x+y}\right)$; P.T., $xu_x + yu_y = 3\tan u$

Soln: $\sin u = \frac{x^2y^2}{x+y}$

$$\Rightarrow \sin u = \frac{x^4(y^2/x^2)}{x(1+\frac{y}{x})} = \frac{x^3(y^2/x^2)}{(1+y/x)}$$

It is a homogenous fnc. with $n=3$.

By euler's theorem,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

$$\Rightarrow x\frac{\partial \sin u}{\partial x} + y\frac{\partial \sin u}{\partial y} = n\,3\sin u$$

$$\Rightarrow x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial y} = 3\sin u$$

$$\Rightarrow \cos u \left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} \right) = 3\sin u$$

$$\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3\tan u$$

Q3 If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$; P.T., $xu_x + yu_y = \sin 2u$

Soln: $\tan u = \frac{x^3+y^3}{x-y}$

$$\Rightarrow \tan u = \frac{x^3}{x}\frac{(1+y^3/x^3)}{(1-y/x)} = x^2\frac{(1+y^3/x^3)}{(1-y/x)}$$

$\Rightarrow$ It is a homogenous fn with $n=2$.

By Euler's theorem,

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

$$\Rightarrow x\frac{\partial \tan u}{\partial x} + y\frac{\partial \tan u}{\partial x} = 2\tan u$$

$$\Rightarrow x\sec^2 u\frac{\partial y}{\partial x} + y\sec^2 u\frac{\partial y}{\partial y} = 2\tan u.$$

$$\Rightarrow \sec^2 u\left[x\frac{\partial u}{\partial x} + y\frac{\partial y}{\partial y}\right] = 2\tan u$$

$$\Rightarrow x\frac{\partial y}{\partial x} + \frac{y\partial u}{\partial y} = \frac{2\tan u}{\sec^2 u} = \frac{2\sin u}{\cos u}\cos^2 u = \sin 2u$$

$$\Rightarrow xu_x + yu_y = \frac{\sin 2u}{2}$$

$$u = e^{\frac{x^3y^3}{x^2+y^2}}$$

Soln $$\log u = \frac{x^3y^3}{x^2+y^2}$$

$$\log u = \frac{x^6(y^3/x^3)}{x^2(1+y^2/x^2)} = \frac{x^4(y^3/x^3)}{(1+y^2/x^2)}$$

$\Rightarrow$ It is a homogenous fnc. with $n = 4$.

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$$

$$\Rightarrow x\frac{\partial(\log u)}{\partial x} + y\frac{\partial(\log u)}{\partial y} = 4\log u$$

$$\Rightarrow x\frac{1}{u}\frac{\partial u}{\partial x} + y\frac{1}{u}\frac{\partial u}{\partial y} = 4\log u$$

$$\Rightarrow x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 4u\log u$$

Total Derivatives

If $u = f(x, y)$, then the total differential of u is defined as $du = \frac{\partial u}{\partial x}\cdot dx + \frac{\partial u}{\partial y}\cdot dy$

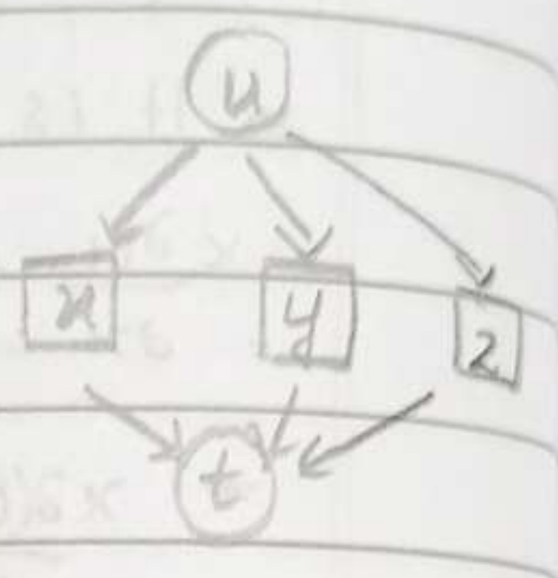
Similarly, for a function $u = f(x, y, z)$, we have,

$$du = \frac{\partial u}{\partial x}\cdot dx + \frac{\partial u}{\partial y}\cdot dy + \frac{\partial u}{\partial z}\cdot dz.$$

(composite function

If $u = f(x, y, z)$, $x = x(t)$, $y = y(t)$, $z = z(t)$, then u is called composite function with single variable. Then the total derivative of u w.r.t. 't' is given by,

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$



If $u = f(x, y, z)$,

$x = x(r, s)$

$y = y(r, s)$

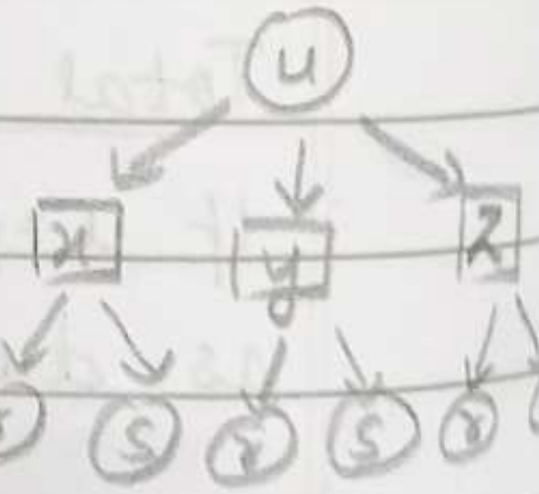
$z = z(r, s)$, then

u is called composite function of two variables r, s

The chain rule for two partial derivatives is given by

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial r}$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial s}$$



Q1. If $u = e^x \sin(yz)$ where $x = t^2$; $y = t - 1$; $z = \frac{1}{t}$

Find $\frac{du}{dt}$ at $t = 1$.

Soln: $\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial u}{\partial z}\cdot\frac{dz}{dt}$

$\Rightarrow \frac{du}{dt} = [\sin(yz)e^x](2t) + (e^x\cos(yz)\cdot z)(1) + (e^x\cos(yz)\cdot y)\left(-\frac{1}{t^2}\right)$

Replace $x = t^2$, $y = t-1$, $z = 1/t$.

$\Rightarrow \frac{dy}{dt} = \left[\sin\left((t-1)\frac{1}{t}\right)\right]e^{t^2} + e^{t^2}\cos\left((t-1)\frac{1}{t}\right)\frac{1}{t} + \left(e^{t^2}\cos\left((t-1)\frac{1}{t}\right)\right)\left(\frac{-1}{t^2}\right)$

substitute $t = 1$,

$= \sin\left\{(1-1)\frac{1}{t}\right\}e^{t^2}(2(1)) + e^1\cos\left\{(1-1)\frac{1}{t}\right\}\frac{1}{t} + e^1\cos\left\{(1-1)\frac{1}{t}\right\}(1-1)(-1)$

$= 0 + e + 0$

$= e$

Q2. If $u = x^2 - y^2$, where $x = 2r - 3s + 4$; $y = -r + 8s - 5$.

Find (i) $\frac{\partial u}{\partial r}$ (ii) $\frac{\partial u}{\partial s}$

Soln: $\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial r}$.

$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial u}{\partial u}\cdot\frac{\partial y}{\partial s}$

Now, $\frac{\partial u}{\partial r} = 2x(2) + (-2y)(-1) = 4x + 2y$

$\frac{\partial u}{\partial s} = 2x(-3) + (-2y)(+8) = -6x - 16y$

$$\frac{\partial u}{\partial r} = 4x + 2y = 4(2r-3s+4) + 2(-r+8s-5).$$
$$= 8r - 12s + 16 - 2r + 16s - 10.$$
$$= 6r + 4s + 6$$

$$\frac{\partial u}{\partial s} = -6x + 6y = -6x - 16y = -6(2r-3s+4) - 16(-r+8s-5)$$
$$= -12r + 18s - 24 + 16r - 128s + 80.$$
$$= 4r - 110s + 56.$$

Q3 If $u = e^x \sin y$; $x = t^2$; $y = t-1$;
Find $\frac{du}{dt}$ at $t = 1$.

Soln:

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$\Rightarrow \frac{du}{dt} = e^x \sin y + e^x \cos y (2t) + e^x \cos y (1)$$

$$\Rightarrow \frac{du}{dt} = e^x \sin y (2t) + e^x \cos y.$$

$$\Rightarrow \frac{du}{dt} = e^x \sin y (2) + e^x \cos y. = e^{t^2} \sin y (2) + e^{t^2}$$
$$= e^{t^2} \sin (t-1)(2) + e^{t^2} \cos (t-1)$$
$$= e^1 \sin (1-1)(2) + e^{t^2} \cos (t-1)$$
$$= e\frac{\pi}{2} e.$$

Q1 If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ PT, $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$. ——— (*)

Soln: Let 'u' be a function of 'r' and 's', i.e., $u = f(r, s)$, then from $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$,

$$r = \frac{y-x}{xy} \quad \text{and} \quad s = \frac{z-x}{xz}$$

$$\Rightarrow r = \frac{1}{x} - \frac{1}{y} \quad \text{and} \quad s = \frac{1}{x} - \frac{1}{z}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial x}$$

$$= \frac{\partial u}{\partial r}\left(\frac{-1}{x^2}\right) + \frac{\partial u}{\partial s}\cdot\left(-\frac{1}{x^2}\right)$$

$$= \frac{-1}{x^2}\left(\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}\right) \quad \text{——— (1)}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial y} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial y}$$

$$= \frac{\partial u}{\partial r}\cdot\left(\frac{1}{y^2}\right) + \frac{\partial u}{\partial s}(0)$$

$$= \frac{1}{y^2}\frac{\partial u}{\partial r} \quad \text{——— (2)}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial z} + \frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial z} = \frac{\partial u}{\partial r}(0) + \frac{\partial u}{\partial s}\left(\frac{1}{z^2}\right)$$

$$= \frac{1}{z^2}\frac{\partial u}{\partial s} \quad \text{——— (3)}$$

Subs ①, ②, ③ in ④

$$\Rightarrow x^2\left(-\frac{1}{x^2}\right)\left(\frac{\partial u}{\partial r}+\frac{\partial u}{\partial s}\right)+y^2\left(\frac{1}{y^2}\frac{\partial u}{\partial r}\right)+z^2\left(\frac{1}{z^2}\frac{\partial u}{\partial s}\right)$$

$$\Rightarrow -\frac{\partial u}{\partial r}-\frac{\partial u}{\partial s}+\frac{\partial u}{\partial r}+\frac{\partial u}{\partial s}$$

$$= 0$$

$\therefore$ Proved.

Q5 If $u = f(xz, y/z)$, P.T., $xu_x - yu_y - zu_z = 0$. —— ⑤

Soln: Let $r = xz$

$$s = \frac{y}{z}$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial x}+\frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial x}$$

$$= \frac{\partial u}{\partial r}\cdot z + \frac{\partial u}{\partial s}\cdot(0)$$

$$= z\frac{\partial u}{\partial r} \quad \text{——} ①$$

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial y}+\frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial y}$$

$$= \frac{\partial u}{\partial r}(0)+\frac{\partial u}{\partial s}\left(\frac{1}{z}\right)$$

$$= \frac{1}{z}\frac{\partial u}{\partial s} \quad \text{——} ②$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial z} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial z}$$

$$= \frac{\partial u}{\partial x}(x) + \frac{\partial u}{\partial s}\left(\frac{-y}{z^2}\right)$$

$$= x\frac{\partial u}{\partial x} - \frac{y}{z^2}\frac{\partial u}{\partial s} \quad \text{——— (3)}$$

substitute (1), (2), (3) in eq. (4),

$$xz\frac{\partial u}{\partial x} - y\frac{1}{2}\frac{\partial u}{\partial s} - z\left[x\frac{\partial u}{\partial x} - \frac{1}{z^2}\frac{\partial u}{\partial s}\right]$$

$$= xz\frac{\partial u}{\partial x} - \frac{y}{2}\frac{\partial u}{\partial s} - zx\frac{\partial u}{\partial x} + \frac{y}{2}\frac{\partial u}{\partial s}$$

$$= \frac{0}{2}$$